

AJAY SINGH

Senior SQL Developer Lead | Database Development | T-SQL Optimization

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Professional Summary

Senior Database Engineer with 20 years of IT experience, including 5 years in US markets, specializing in T-SQL development, database architecture, and enterprise reporting. Proven expertise in SQL Server, SSRS, SSIS, and Power BI. Skilled in performance tuning, unit testing, and agile development practices. Recently led the design and implementation of a **custom T-SQL unit testing framework** integrated with CI tools like TEAMCITY. Proficient in debugging with Visual Studio, leveraging GitHub Copilot Agent in VS Code to accelerate test case generation. Adept at using **SQL Profiler Trace** for functional flow analysis, and experienced with JIRA, Bitbucket, GitHub, and SourceTree in collaborative development environments. Recognized for optimizing query performance and delivering scalable, high-quality BI solutions.

Technical Skills

- **Forecasting and Modeling:** Expertise in developing forecast models using SQL and BI tools to analyze trends and predict demand patterns.
- **Risk Mitigation:** Skilled in identifying risks, conducting impact assessments, and implementing mitigation strategies to ensure data integrity and operational stability.
- **Platform and Data Issue Resolution:** Adept at troubleshooting and resolving platform and data-related issues, reducing downtime by implementing efficient query and ETL optimizations.
- **Process Documentation:** Extensive experience in creating and maintaining process documentation, including SOPs, ensuring compliance and operational clarity.
- **Programming & Scripting:** T-SQL, SQL, PL/SQL, DAX, Dynamic SQL, GitHub Copilot Agent (VS Code)
- **Databases & Tools:** Microsoft SQL Server (2016/2019), SSMS, SQL Profiler, SQL Trace, Azure SQL
- **Database Development:** Stored Procedures, Functions, Views, Triggers, Performance Tuning, Query Optimization, Database Design, Data Modeling, Unit Testing Framework (Custom T-SQL based)
- **DevOps & Source Control:** GitHub, Bitbucket, SourceTree, TEAMCITY (CI Integration), Jenkins
- **Debugging & IDEs:** Visual Studio, VS Code, SQL Debugger, GitHub Copilot Agent
- **Project & Agile Tools:** JIRA (Agile/Scrum Task Tracking), Confluence, Technical Documentation, Sprint Planning

Business Intelligence and Visualization

- **Dashboards:** Power BI, Oracle BIEE, KPI Visualizations, DAX, Data Transformation
- **Data Analysis:** Forecasting Models, Data Integration, Trend Analysis, Risk Mitigation
- **Report Development:** Migration from Crystal Reports to SSRS, Advanced SQL-Based Reports

Process and Performance Optimization

- **Performance Tuning:** Query Optimization, Indexing Strategies, Database Scalability Improvements
- **Automation:** Reporting Process Automation, CI/CD Pipelines (Jenkins), Workflow Optimization

Soft Skills

- Business Acumen, Attention to Detail, Collaboration, Effective Communication, Presentation Skills, Organizational Skills, Multitasking, Adaptability
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Professional Experience

Project: Custom Database Unit Testing Framework & CI Integration

Launch Consulting Group, Hyderabad

Client: Healthedge

Dec 2024 – Present

- Designed and implemented a **custom unit test framework** using T-SQL to validate stored procedures, functions, and views.
- Utilized **SQL Server Profiler Trace** to map procedures involved in functional flows, enabling precise test coverage.
- Enabled **dynamic test data** generation for each test case, supporting robust and repeatable testing.
- Integrated **GitHub Copilot Agent with VS Code**, improving efficiency in identifying test cases and automating code patterns.
- Leveraged **Visual Studio debugging tools** to trace complex SQL execution and validate results.
- Used **Bitbucket and GitHub** for source control; collaborated with the team via **SourceTree, GitBash** for streamlined code check-ins.
- Managed tasks in an **Agile environment using JIRA**, ensuring sprint alignment and traceability.
- Familiar with **TEAMCITY** to execute automated unit tests across multiple environments.

Senior Technical Business Analyst

TechTriad, Hyderabad

Client: Prudential Financial (NJ)

Sep 2021 – Oct 2024

- Partnered with business units to assess reporting needs, improving alignment with goals and enhancing data-driven decision-making.
- Designed and developed SQL-based dashboards and reports using Oracle BI, providing actionable insights and boosting operational efficiency.
- Documented technical requirements and ensured clear specifications for development and testing phases, increasing process clarity.
- Conducted User Acceptance Testing (UAT), creating detailed scripts to validate accuracy and usability of dashboards and reports.
- Improved reporting system workflows through SQL query optimization, reducing data retrieval time by 25%.

Programmer Analyst

The Evolvers Group, Texas

Client: Energy Transfer / City of Arlington (TX)

June 2016 – June 2021

- Led the migration of 100+ Crystal Reports to SSRS, implementing a robust strategy that ensured seamless transitions without data loss.
- Developed and optimized SSRS reports (drill-through, tabular, and matrix), reducing report generation time by 40%.
- Enhanced SQL Server performance through indexing strategies and query tuning, increasing database efficiency by 30%.
- Trained team members on SQL query optimization and SSRS best practices, strengthening technical capabilities within the team.
- Collaborated with stakeholders to gather requirements, ensuring all reporting solutions met business objectives.

Manager - Development and Analytics

C3i HealthCare Connections, Hyderabad

Jan 2005 – March 2016

- Led the design, development, and maintenance of Crystal Reports and SQL Server-based reporting solutions, automating manual processes to reduce reporting time by up to 40%.
 - Migrated legacy Crystal Reports to Interaction Level Queries (ILQ) and SSRS, enhancing scalability and improving data accuracy.
 - Spearheaded SharePoint migration from 2010 to 2013, optimizing portal functionality and ensuring seamless user adoption.
 - Managed custom tool enhancements (eHOC and QC Tool) and integrated ETL processes using SSIS, addressing scalability and reliability challenges.
 - Collaborated with clients and stakeholders to gather requirements, translating them into optimized reporting solutions aligned with business objectives.
 - Trained and mentored junior team members on Crystal Reports, SQL query optimization, and report development best practices, building a technically proficient team.
 - Developed and deployed various .NET-based client portals and internal tools, contributing to the organization's digital transformation initiatives.
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Education

Master of Computer Applications (MCA)

Vinayak Mission Research Foundation, Chennai, India

Bachelor of Commerce

Tilka Manjhi Bhagalpur University, Bhagalpur, India

Key Achievements

- Developed a scalable, in-house unit testing framework using T-SQL to automate testing targeted for over 4000+ stored procedures.
- Adopted GitHub Copilot Agent in VS Code to identify and accelerate test case development. Improved defect detection in pre-UAT cycles by 40% and reduced manual QA effort by 60%.
- Achieved a 30% improvement in data processing times through database performance tuning and query optimization.
- Spearheaded data migration projects, utilizing SSIS for ETL processes that enhanced data flow and streamlined reporting.
- Enhanced reporting efficiency by developing parameterized and matrix SSRS reports, providing stakeholders with real-time, actionable insights.